

tRNA m7G46 Methylation: The TrmB Module of *Pseudomonas putida* KT2440 (UPA00989)

A commissioned review-style synthesis

1. Executive Summary

The *Pseudomonas putida* KT2440 UniPathway module **UPA00989** describes a single, chemically simple, but biologically consequential reaction: the S-adenosyl-L-methionine (SAM)–dependent transfer of a methyl group to the **N7 atom of guanine-46** in the variable loop of transfer RNA, producing **7-methylguanosine (m7G46)**. The reaction is carried out by **TrmB**, the tRNA (guanine(46)-N7)-methyltransferase encoded by **PP_5103** — the *P. putida* ortholog of *E. coli* YggH and of the biochemically characterized *P. aeruginosa* TrmB. As a biological system, the module is best understood not as a self-contained pathway but as **one obligatory catalytic step that behaves as a node in a larger, cooperative tRNA-modification and quality-control network**.

Three ideas organize this review. First, the **chemistry and catalytic scaffold are ancient and near-universal**: TrmB is a Class I (Rossmann-fold) methyltransferase whose catalytic domain traces back toward the Last Universal Common Ancestor (LUCA), and m7G46 is among the most broadly conserved tRNA modifications across bacteria, archaea, and eukaryotes. Second, the **active assembly is lineage-variable**: TrmB is a monomer in *E. coli* and (by orthology) *Pseudomonas*, a homodimer in *Bacillus subtilis*, and an obligate heterodimer in eukaryotes (yeast Trm8–Trm82; human METTL1–WDR4). Third, the **physiological importance of m7G46 is conditional**: it is largely dispensable under permissive laboratory conditions but becomes critical under thermal and oxidative stress, where it stabilizes tRNA, supports the deposition of other modifications, and tunes codon-biased translation.

For the *P. putida* module specifically, the strongest direct evidence comes from the close relative *P. aeruginosa*, where TrmB forms m7G46 on tRNAs bearing G46 and where its loss impairs translation of phenylalanine- and aspartate-enriched messages (including catalases)

and causes hydrogen-peroxide sensitivity. Direct enzymatic characterization of PP_5103 itself is limited, so the model presented here is **orthology-based but well-triangulated** across enzymology, structural biology, and comparative genomics. Throughout, the tRNA m7G46 system must be kept **distinct** from the mechanistically separate rRNA m7G (Bud23–Trm112 / RsmG) and internal mRNA m7G pathways, and its dedicated accessory subunit (Trm82/WDR4) must be distinguished from the promiscuous methyltransferase activator Trm112/TRMT112.

2. Definition and Biological Boundaries

2.1 What the system is

UPA00989 comprises exactly **one reaction**:



The molecular player is **TrmB (PP_5103)**, a stand-alone bacterial enzyme. Its substrate is the subset of cellular tRNAs that carry an unmodified guanosine at position 46, which lies in the **variable loop** (the connector between the anticodon stem and the T-arm). In *P. aeruginosa*, 23 tRNA species with G46 were identified as candidate substrates. The methyl group is delivered from SAM, and the enzyme belongs to the **Rossmann-fold methyltransferase (RFM / Class I)** superfamily.

2.2 What lies *outside* the boundary

Because "m7G" appears in several unrelated contexts, careful boundary-drawing is essential. The following are frequently conflated with tRNA m7G46 but are **separate systems**:

- **rRNA m7G.** In eukaryotes, 18S rRNA position 1575 is methylated by **Bud23–Trm112** (human WBSCR22–TRMT112) during 40S ribosome biogenesis; the bacterial counterpart on 16S rRNA (m7G527) is installed by **RsmG (GidB)**. These use different writers, different substrates, and different biology from tRNA m7G46.
- **mRNA m7G.** The 5' **cap** m7G and recently described **internal** mRNA m7G marks are distinct epitranscriptomic features, again with separate machinery and readout.
- **The Trm112 activator network.** Trm112 (human TRMT112) is a small allosteric activator that partners with a *set* of translation-related methyltransferases — Bud23 (rRNA m7G),

Trm9, Trm11, Mtq2 (eRF1 methylation) — and is **not** the partner of the tRNA m7G46 writer. The dedicated m7G46 partner in eukaryotes is **Trm82/WDR4**.

2.3 Competing definitions

The main definitional tension is "**single reaction**" vs. "**network node**." UniPathway and enzyme databases treat UPA00989 as one isolated catalytic step (the correct atomic definition). The functional-genomics and physiology literature, however, shows that m7G46 exerts its effects **only in the context of other modifications** (e.g., Gm18, m1G37, m5U54) and of tRNA-decay surveillance. Both framings are valid at different levels of description; this review adopts the reaction as the *boundary* while treating the network as the *context*.

3. Mechanistic Overview

3.1 The best current model

The consensus sequence of events is:

1. **Substrate selection.** TrmB binds a tRNA presenting G46 in an appropriately sized variable loop. Recognition depends critically on the **variable-loop size** and on the **T-stem**, rather than on the full three-dimensional L-shape of mature tRNA.
2. **SAM binding.** SAM docks in the Rossmann-fold catalytic domain.
3. **Methyl transfer.** The methyl group is transferred to N7 of G46, yielding a positively charged 7-methylguanosine and releasing SAH.
4. **Release and downstream coupling.** The modified tRNA re-enters the pool, where m7G46 contributes to structural stability and enables/supports other modification events.

The reaction is **SN2-type methyl transfer** typical of Class I MTases: SAM as methyl donor, direct transfer to the nucleobase nitrogen, no covalent enzyme intermediate required. Notably, **N7 methylation does not alter Watson–Crick pairing** (N7 faces the major groove), consistent with a primarily **structural/stabilizing** rather than decoding-altering role at this position.

3.2 Obligatory, conditional, and accessory elements

Element	Status	Basis
SAM as methyl donor	Obligatory	Class I RFM chemistry
G46 in a variable loop of adequate size	Obligatory (substrate)	Recognition depends on variable-loop size
T-stem contacts	Obligatory (recognition)	Most important recognition element in <i>A. aeolicus</i>
Full L-shaped tertiary fold	Dispensable	Truncated substrates still methylated
Accessory subunit (Trm82/WDR4)	Conditional (lineage-specific)	Required in eukaryotes, absent in <i>E. coli</i> / <i>Pseudomonas</i>
Homodimerization	Accessory (lineage-specific)	Seen in <i>B. subtilis</i> , not <i>E. coli</i>
m7G46 itself for baseline growth	Conditional/ accessory	Dispensable at permissive temperature; required under stress

3.3 Molecular assemblies carrying out each step

The single catalytic step is executed by different physical assemblies across life (see Section 5), but the **catalytic core is a conserved RFM domain** in every case. In bacteria the enzyme acts alone (or as a homodimer in Bacilli); in eukaryotes it requires a catalytically silent partner that stabilizes the writer and sharpens substrate discrimination.

4. Major Molecular Players and Active Assemblies

4.1 TrmB (PP_5103) — the catalytic writer

TrmB is the SAM-dependent tRNA (guanine(46)-N7)-methyltransferase at the center of the module. Direct enzymatic characterization in *P. aeruginosa* — the nearest well-studied relative of *P. putida* KT2440 — established that *trmB* encodes the enzyme forming m7G46 in the variable loop ([PMID: 31428787](#)): "*the P. aeruginosa trmB gene encodes a tRNA guanine (46)-N7-methyltransferase that catalyzes the formation of m7G46 in the tRNA variable loop.*"

Crystallographic work on the *E. coli* and *B. subtilis* enzymes places TrmB firmly in the Rossmann-fold class (PMID: 16600901): *"It reveals a unique variant of the Rossmann-fold methyltransferase (RFM) structure, with the N-terminal helix folded on the opposite site of the catalytic domain."* This is **Finding F001**.

4.2 Substrate recognition module

TrmB does not require an intact L-shaped tRNA. Work on *Aquifex aeolicus* TrmB (PMID: 15358762) showed that *"the enzyme specificity was critically dependent on the size of the variable loop"* and that *"the L-shaped tRNA structure is not required for methyl acceptance activity."* The T-stem emerges as the single most important recognition element. This defines the **local variable-loop/T-stem architecture**, not global tertiary structure, as the recognition determinant (**Finding F004**), and it distinguishes TrmB from elbow-modifying enzymes (e.g., TruB, TrmA) that read the tertiary core. Consistent with a minimal catalytic requirement, the isolated **core domain** of *A. aeolicus* TrmB retains methyl-transfer activity (PMID: 17150909), and the *E. coli* enzyme's tRNA-binding mechanism has been dissected biochemically (PMID: 36933808).

4.3 Accessory subunits — the eukaryotic heterodimer

In eukaryotes the writer is catalytically inert on its own. Yeast **Trm8–Trm82** is active only when both subunits are **co-translated** (PMID: 18029735): *"Active Trm8-Trm82 heterodimer was only synthesized under conditions, in which both Trm8 and Trm82 mRNAs were co-translated."* The human orthologs are **METTL1–WDR4**, whose structures and catalytic mechanism have been solved (PMID: 36599982). Importantly, **Trm82/WDR4 is dedicated to the m7G46 writer** and is *distinct* from the general activator **Trm112/TRMT112**, which partners with a different client set (PMID: 28134793: *"The last complex formed between Trm112 and Bud23 proteins modifies 18S rRNA and participates in the 40S biogenesis pathway"*; PMID: 30010922: *"Trm112 is a general partner for methyltransferases in all living organisms"*). This is the substance of **Findings F003 and F006**.

4.4 The network partners (context, not core)

m7G46 does not act in isolation. In *Thermus thermophilus*, m7G46 supports the deposition of **Gm18** and **m1G37** and is embedded in a modification network (PMID: 19934251). The related folate/FAD-dependent enzyme **TrmFO** (m5U54) participates in the same class of temperature-tuned networks (PMID: 27238446). These enzymes provide context; they are not members of UPA00989 itself.

5. Evolutionary and Cell-Biological Variation

5.1 Quaternary-structure variation across lineages

The single conserved reaction is carried out by different assemblies:

Lineage	Enzyme	Active assembly	Accessory subunit
<i>E. coli</i> (γ -proteobacteria)	TrmB (YggH)	Monomer	None
<i>Pseudomonas</i> spp.	TrmB (PP_5103 ortholog)	Monomer (by orthology)	None
<i>Bacillus subtilis</i> (Firmicutes)	TrmB	Homodimer	None
<i>Aquifex aeolicus</i> (thermophile)	TrmB	Monomer; extended C-terminus	None
<i>S. cerevisiae</i>	Trm8	Obligate heterodimer	Trm82
<i>Homo sapiens</i>	METTL1	Obligate heterodimer	WDR4

The homodimerization of *B. subtilis* TrmB is explicitly lineage-specific (PMID: 16600901): "unlike EcTrmB, BsTrmB is shown here to be dimeric both in the crystal and in solution." The eukaryotic heterodimer recognizes tRNA more stringently than the bacterial monomer. This spread — monomer → homodimer → obligate heterodimer — is a textbook example of **conserved chemistry with elaborated assembly** (Finding F003).

5.2 Conservation and origin

The **catalytic RFM scaffold is ancient**. Comparative and phylogenetic analyses of RNA methyltransferases indicate that several MTase catalytic domains predate the divergence of the domains of life (PMID: 40508070): "indicating their ancient origin in the Last Universal Common Ancestor (LUCA)." m7G46 is one of the most broadly distributed tRNA modifications. The picture is therefore of an **ancient catalytic core** onto which **later, lineage-specific elaborations** (auxiliary domains, oligomerization, obligate partner subunits) were layered

(Finding F007). For understanding the *ancestral* role, the **bacterial monomeric enzymes (E. coli, Pseudomonas) are the best representatives**, because they retain the minimal writer without the eukaryote-specific heterodimeric apparatus.

5.3 Physiological-state and cell-type variation

Because N7 methylation is chemically identical everywhere, variation is expressed mainly through **when and how much** the mark matters:

- **Thermophiles / high temperature.** In *T. thermophilus*, m7G46 is required for viability at high temperature and supports the modification network; loss lowers tRNA melting temperature and depresses protein synthesis above 70 °C (PMID: 19934251).
- **Oxidative stress.** In *P. aeruginosa*, m7G rises upon H2O2 exposure and TrmB loss impairs catalase translation (PMID: 31428787).
- **Eukaryotic disease states.** METTL1/WDR4-mediated tRNA m7G is dysregulated in multiple cancers, where it modulates translation of oncogenic messages (PMID: 34371184, PMID: 42010934) — a eukaryote-specific elaboration of the same core mark, retained here only as context.

5.4 Alternative routes

There is **no known alternative enzyme** that installs m7G46; the reaction is monopolized by the TrmB/METTL1 family. "Alternative routes" exist only at the level of the *network outcome* (tRNA stability/translation), which is buffered by other modifications and, when it fails, by tRNA-decay surveillance.

6. Constraints, Dependencies, and Failure Modes

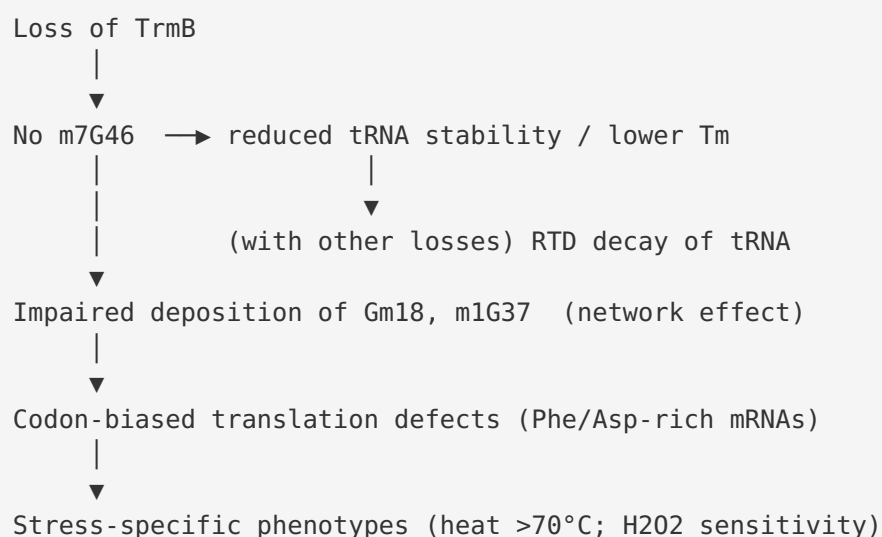
6.1 Ordering and dependency constraints

- **SAM must be bound before transfer** (obligate co-substrate ordering).
- **G46 must be presented in a correctly sized variable loop** for recognition; the T-stem is required for productive binding.
- **In the modification network, m7G46 acts upstream of / supports other modifications.**
In *T. thermophilus*, m7G46 supports introduction of Gm18 and m1G37 (PMID: 19934251):
"the m(7)G46 modification is required for cell viability at high temperatures via a tRNA

modification network, in which the m(7)G46 modification supports introduction of other modifications." This is the clearest ordering constraint in the system (**Finding F002**).

6.2 Failure modes

The dominant failure mode of losing m7G46 is **tRNA destabilization and surveillance-mediated decay**. In yeast, mature tRNA-Val(AAC) lacking both m7G and m5C is degraded by the **rapid tRNA decay (RTD)** pathway (Rat1/Xrn1/Met22) ([PMID: 18443146](#)): "*mature tRNA(Val(AAC)) lacking 7-methylguanosine and 5-methylcytidine is rapidly degraded and deacylated at 37 degrees C in a trm8-Delta trm4-Delta strain.*" The downstream consequence is **codon-biased translational failure**: in *P. aeruginosa*, TrmB loss selectively impairs translation of Phe- and Asp-enriched mRNAs ([PMID: 31428787](#)): "*loss of trmB had a strong negative effect on the translation of Phe- and Asp-enriched mRNAs*" (**Finding F005**).



6.3 What is ruled out

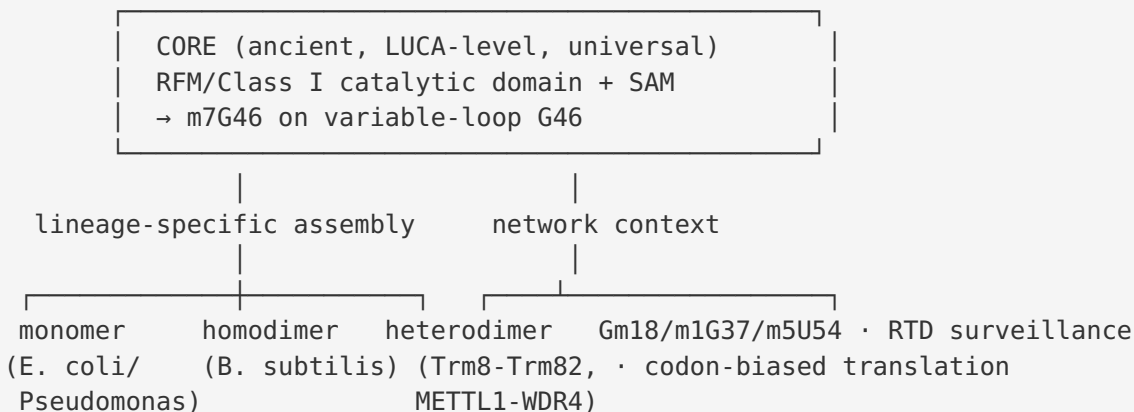
- **Full L-shape requirement is ruled out:** truncated substrates lacking tertiary structure are still methylated ([PMID: 15358762](#)).
- **Baseline essentiality is ruled out:** at permissive temperature, m7G46 loss is tolerated; essentiality is conditional on stress.
- **Shared machinery with rRNA/mRNA m7G is ruled out:** those marks are installed by distinct writers (Bud23–Trm112 / RsmG; cap and internal mRNA enzymes) ([PMID: 25489090](#)).

7. Mechanistic Model / Consolidated Interpretation

Integrating eight findings across 28 papers yields a coherent model (**Finding F008**):

- **Core identity.** TrmB (PP_5103 ortholog) catalyzes SAM-dependent m7G46 in the tRNA variable loop ([PMID: 31428787](#)).
- **Enzyme class and recognition.** It is a Rossmann-fold Class I MTase that reads the variable-loop/T-stem, not the full L-shape ([PMID: 16600901](#), [PMID: 15358762](#)).
- **Conditional importance.** Dispensable at baseline; required under thermal/oxidative stress; acts in a cooperative modification network and gates codon-biased translation ([PMID: 19934251](#), [PMID: 31428787](#)).
- **Failure surveillance.** Loss (with other modification losses) triggers rapid tRNA decay ([PMID: 18443146](#)).
- **Evolution.** The catalytic core is LUCA-ancient, while accessory subunits (Trm82/WDR4, distinct from Trm112) and oligomerization are lineage-specific elaborations ([PMID: 40508070](#), [PMID: 28134793](#), [PMID: 18029735](#)).

In one sentence: **UPA00989 is one ancient, conserved reaction that behaves as a stress-tuned node in the tRNA-modification and quality-control network, executed by a catalytic scaffold that is universal even though its active assembly is lineage-variable.**



8. Evidence Base

PMID	Title (abbreviated)	Role in this review
31428787	<i>TrmB, a tRNA m7G46 methyltransferase, in H2O2 resistance (P. aeruginosa)</i>	Primary anchor for enzyme identity in Pseudomonas; codon-biased translation & oxidative-stress phenotype (F001, F005)
16600901	<i>Crystal structure of B. subtilis TrmB</i>	Rossmann-fold classification; lineage-specific homodimerization (F001, F003)
15358762	<i>Substrate tRNA recognition by A. aeolicus TrmB</i>	Variable-loop/T-stem recognition; L-shape dispensable (F004)
19934251	<i>m7G46 required for viability at high temperature (T. thermophilus)</i>	Network node; supports other modifications; stress essentiality (F002, F008)
18443146	<i>Degradation of hypomodified tRNAs (RTD pathway)</i>	Failure mode: rapid tRNA decay on m7G loss (F002)
18029735	<i>Yeast Trm8–Trm82 heterodimer assembly</i>	Obligate co-translational heterodimer in eukaryotes (F003)
28134793	<i>Trm112, activator of translational MTases</i>	Distinguishes Trm112 (rRNA m7G/Bud23) from m7G46 partner (F006)
30010922	<i>Trm112-methyltransferase holoenzymes, archaea vs eukaryotes</i>	Trm112 as broad multi-client activator (F006)
25489090	<i>Bud23–Trm112 and 18S rRNA m7G1575</i>	Separates rRNA m7G from tRNA m7G46 (F005)
40508070	<i>Evolutionary origins of 2'-O-MTases / RNA-MTases</i>	LUCA-level origin of RFM catalytic core (F007)
17150909	<i>Core domain of A. aeolicus TrmB has methyl-transfer activity</i>	Minimal catalytic core suffices; thermophile C-terminal variation
36933808	<i>Molecular mechanism of tRNA binding by E. coli TrmB</i>	Biochemistry of bacterial TrmB–tRNA binding
36599982	<i>Structures and mechanisms of METTL1–WDR4</i>	Eukaryotic heterodimer structure (context)
27238446	<i>TrmFO regulates other modifications at low temperature</i>	Modification-network context

PMID	Title (abbreviated)	Role in this review
34371184	<i>METTL1/WDR4 tRNA m7G in cancer</i>	Eukaryotic disease-state elaboration (context)

How the evidence fits together. The *P. aeruginosa* study (PMID: [31428787](#)) is the load-bearing anchor for the *Pseudomonas* module because it is the closest relative of *P. putida* with direct enzymatic and phenotypic data. Structural biology (PMID: [16600901](#), PMID: [36933808](#)) fixes the enzyme class and recognition logic. Thermophile genetics (PMID: [19934251](#), PMID: [17150909](#)) reveal the network and conditional-essentiality behavior. Eukaryotic and comparative studies (PMID: [18029735](#), PMID: [28134793](#), PMID: [40508070](#)) establish the ancient core and lineage-variable assembly. No paper in the set contradicts the core model; the main tension is one of *emphasis* (single reaction vs. network node).

9. Controversies and Open Questions

- 1. Direct characterization of PP_5103.** The *P. putida* KT2440 enzyme itself has not been directly characterized enzymatically; the assignment rests on strong orthology to *P. aeruginosa* TrmB and *E. coli* YggH. The full substrate set, kinetics, and stress phenotypes of PP_5103 remain to be measured directly.
- 2. Single reaction vs. network node.** The literature productively disagrees on framing. Atomically, UPA00989 is one reaction; physiologically, m7G46's effects are inseparable from the surrounding modification network and from tRNA-decay surveillance. Both are correct at different levels.
- 3. Mechanism of "network support."** How m7G46 *facilitates* deposition of Gm18 and m1G37 is not resolved at the structural level — whether through direct conformational priming, altered folding kinetics, or thermal stabilization that keeps substrates in a competent state.
- 4. Cross-organism generalization.** Much mechanistic detail comes from thermophiles (*T. thermophilus*, *A. aeolicus*) and from eukaryotes (yeast, human). Extrapolating quantitative importance to a mesophilic soil bacterium like *P. putida* should be done cautiously; the *qualitative* biochemistry is robust, but the *magnitude* of stress phenotypes may differ.

5. **Accessory-subunit logic.** Why eukaryotes require an obligate heterodimer while bacteria do not is not fully explained. The partner (Trm82/WDR4) appears to stabilize the writer and sharpen specificity, but the evolutionary driver for making it *obligate* remains an open question.
-

10. Limitations and Knowledge Gaps

1. **Orthology-based inference.** The strongest limitation is that **PP_5103 has not been directly characterized**. All claims about the *P. putida* enzyme rest on orthology to *P. aeruginosa* TrmB and *E. coli* YggH. While this inference is well-supported, the exact substrate set, kinetics, and knockout phenotype of *P. putida* KT2440 TrmB remain experimentally unverified.
 2. **Cross-organism mixing.** Mechanistic depth comes disproportionately from thermophiles and eukaryotes. The *quantitative* magnitude of stress phenotypes in a mesophilic soil bacterium is uncertain, even if the qualitative biochemistry is robust.
 3. **Network mechanism unresolved.** *How* m7G46 supports downstream modifications (Gm18, m1G37) is not mechanistically defined.
 4. **Regulation of the writer.** Whether and how *trmB* expression or TrmB activity is regulated in *P. putida* under stress is unknown.
 5. **No structural data for the Pseudomonas enzyme.** All available structures are from *E. coli*, *B. subtilis*, *A. aeolicus*, or the eukaryotic heterodimer.
-

11. Proposed Follow-up Experiments / Actions

1. **Direct biochemical characterization of PP_5103.** Express and purify *P. putida* KT2440 TrmB; confirm SAM-dependent m7G46 activity in vitro on defined tRNA substrates; measure kinetic parameters and quaternary state (monomer vs. dimer by SEC-MALS).
2. **Substrate mapping in *P. putida*.** Use tRNA-seq / mass spectrometry or m7G-specific sequencing to enumerate the G46-bearing tRNAs actually methylated in vivo and quantify occupancy.

3. **Clean knockout phenotyping.** Construct a ΔPP_5103 strain; assay growth under thermal stress and oxidative stress (H₂O₂, paraquat), and test the predicted catalase-translation / peroxide-sensitivity phenotype seen in *P. aeruginosa*.
4. **Network coupling test.** In the knockout, quantify secondary hypomodification (e.g., changes at other tRNA positions) to test whether the *T. thermophilus* "modification-network" behavior extends to a mesophile.
5. **tRNA-stability / decay assay.** Test whether m7G46 loss (alone or combined with another modification deficiency) accelerates tRNA turnover in *P. putida*, probing whether an RTD-like surveillance operates.
6. **Structural determination.** Solve a *Pseudomonas* TrmB structure (crystallography or cryo-EM) to confirm the RFM fold and map the variable-loop/T-stem recognition surface in this lineage.
7. **Codon-usage / translation profiling.** Ribosome profiling of wild-type vs. ΔPP_5103 to test for codon-biased translational defects analogous to the Phe/Asp-enriched-message effect in *P. aeruginosa*.

12. Key References

- *TrmB, a tRNA m7G46 methyltransferase, plays a role in hydrogen peroxide resistance and positively modulates the translation of katA and katB mRNAs in Pseudomonas aeruginosa.* PMID: 31428787
- *Crystal structure of Bacillus subtilis TrmB, the tRNA (m7G46) methyltransferase.* PMID: 16600901
- *Substrate tRNA recognition mechanism of tRNA (m7G46) methyltransferase from Aquifex aeolicus.* PMID: 15358762
- *N7-Methylguanine at position 46 (m7G46) in tRNA from Thermus thermophilus is required for cell viability at high temperatures through a tRNA modification network.* PMID: 19934251
- *Degradation of several hypomodified mature tRNA species in Saccharomyces cerevisiae is mediated by Met22 and the 5'-3' exonucleases Rat1 and Xrn1.* PMID: 18443146
- *Hetero subunit interaction and RNA recognition of yeast tRNA (m7G46) methyltransferase synthesized in a wheat germ cell-free translation system.* PMID: 18029735

- *Trm112, a Protein Activator of Methyltransferases Modifying Actors of the Eukaryotic Translational Apparatus.* [PMID: 28134793](#)
- *Evolutionary insights into Trm112-methyltransferase holoenzymes involved in translation between archaea and eukaryotes.* [PMID: 30010922](#)
- *Structural and functional studies of Bud23-Trm112 reveal 18S rRNA N7-G1575 methylation occurs on late 40S precursor ribosomes.* [PMID: 25489090](#)
- *Evolutionary Origins and Functional Diversification of 2'-O-Methyltransferases: Insights from Phylogenetic and Structural Analysis.* [PMID: 40508070](#)
- *The core domain of Aquifex aeolicus tRNA (m7G46) methyltransferase has the methyl-transfer activity to tRNA.* [PMID: 17150909](#)
- *Molecular mechanism of tRNA binding by the Escherichia coli N7 guanosine methyltransferase TrmB.* [PMID: 36933808](#)
- *Structures and mechanisms of tRNA methylation by METTL1-WDR4.* [PMID: 36599982](#)

Prepared as a commissioned review synthesis of the P. putida KT2440 UPA00989 (TrmB / m7G46) module. Direct characterization of PP_5103 is limited; the model is orthology-based and triangulated across enzymology, structural biology, and comparative genomics. Claims are anchored to primary literature, and uncertainty is flagged where extrapolation across organisms or assay systems is required.
